1. [7 pts] Evaluate the Integrate: $\int (x^2 + 1) \cos x \; dx$

$(x^2+1)\sin(x) + 2x\cos(x) - 2\sin(x) + C$

work on paper

+7

2. [5 pts] Evaluate the Integrate: $\int_0^1 32xe^{4x} dx$

$6e^4 + 2$

work on paper

+4

3. [5 pts] Evaluate the Integrate: $\int \cot^3 x \csc^5 x \, dx$

$$-\frac{\csc^7(x)}{7} + \frac{\csc^5(x)}{5} + c$$

+5

work is on paper

4. [7 pts] Evaluate the Integrate: $\int \tan^3 x \, dx$

$$\frac{\tan^2(x)}{2} + \ln|\cos x| + c$$

work on paper

+7

5. [7 pts]

5. [7 pts] Evaluate the Integrate: $\int \sin^2(2x)\cos^2(2x)\, dx$

$u = 2x$
$du = 2dx$
$\frac{du}{2} = dx$

$\sin^2(x) = \frac{1}{2} - \frac{1}{2}\cos(2x)$
$\cos^2(x) = \frac{1}{2} + \frac{1}{2}\cos(2x)$

$$\int \cancel{\sin^2(u)\cos^2(u) \cdot \frac{du}{2}}$$

+2

$$\frac{1}{2}\int \sin^2(u)\cos^2(u)\, du$$

$$\frac{1}{2}\int \left(\frac{1}{2} - \frac{1}{2}\cos(\overset{2u}{\cancel{2x}})\right)\left(\frac{1}{2} + \frac{1}{2}\cos(\overset{2u}{\cancel{2x}})\right)$$

To many errors!
Mixed u & x.

$$\frac{1}{2}\int \frac{1}{4} + \cancel{\frac{1}{4}\cos(2x)} - \cancel{\frac{1}{4}\cos(2x)} - \frac{1}{4}\cos^2(4x)$$

$$\frac{1}{2}\int \frac{1}{4} - \frac{1}{4}\cos^2(4x) \Rightarrow \frac{1}{2}\int \frac{1}{4} - \frac{1}{4}\int \cos^2(4x)\, dx$$

$u = 4x$
$du = 4dx$
$\frac{du}{4} = dx$

$$\frac{1}{2}\left[\int \frac{1}{4} - \frac{1}{16}\int \cos^2(u)\, du\right]$$

$$\frac{1}{8}x - \frac{1}{16}\int \cos^2(u)\, du \Rightarrow \frac{1}{8}x - \frac{1}{16}\int \frac{1}{2} + \frac{1}{2}\cos(2u)$$

$$\frac{1}{8}x - \frac{1}{32}\int 1 + \frac{1}{2}\cos(2u) \Rightarrow \frac{1}{8}x - \frac{1}{32}x + \frac{1}{2}\int \cos(2u)$$

$u = 8x$
$du = 8dx$
$\frac{du}{8} = dx$

$$\frac{3}{32}x + \frac{1}{2}\int \cos(8x)\, dx$$

$$\boxed{\frac{3}{32}x + \frac{1}{16}\sin(8x) + C}$$

$$\left(\frac{1}{2} - \frac{1}{2}\cos 4x\right)\left(\frac{1}{2} + \frac{1}{2}\cos 4x\right)$$

$$\int \left(\frac{1}{4} - \frac{1}{4}\cos^2 4x\right)dx = \int \frac{1}{4} - \frac{1}{4}\left(\frac{1}{2} + \frac{1}{2}\cos(8x)\right)$$

$$= \int \left(\frac{1}{8} - \frac{1}{8}\cos(8x)\right) dx \quad u = 8x \quad \ldots$$

$\tan^2\theta + 1 = \sec^2\theta$

~~6.~~ [7 pts] Evaluate the Integral: $\int \frac{1}{\sqrt{x^2-9}}\,dx$

$x = 3\sec\theta \qquad dx = 3\sec\theta\tan\theta\,d\theta$

$$\int \frac{1}{\sqrt{(3\sec\theta)^2-9}} \cdot 3\sec\theta\tan\theta\,d\theta \Rightarrow \int \frac{1}{\sqrt{9\sec^2\theta-9}} \cdot 3\sec\theta\tan\theta\,d\theta$$

$$\int \frac{1}{\sqrt{9(\sec^2\theta-1)}} \cdot 3\sec\theta\tan\theta\,d\theta \qquad \int \frac{1}{\sqrt{9\tan^2\theta}} \cdot 3\sec\theta\tan\theta\,d\theta$$

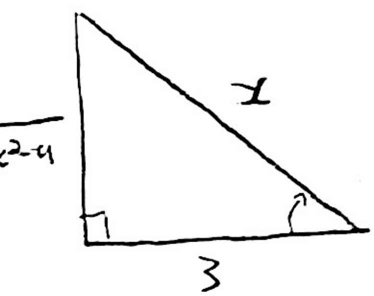
$$\int \frac{1}{\cancel{3}\cancel{\tan\theta}} \cdot \cancel{3}\sec\theta\,\cancel{\tan\theta}\,d\theta \checkmark \Rightarrow \int \sec\theta\,d\theta \checkmark$$

+7

$$\ln|\sec\theta + \tan\theta| + C \checkmark$$

$\sec\theta = \frac{x}{3} \quad \frac{hyp}{adj}$

$\tan\theta = \frac{\sqrt{x^2+9}}{3}$



✓

SOH-CAH-TOA

$\frac{1}{\cos\theta}, \frac{1}{\frac{adj}{hyp}} = \frac{hyp}{adj}$

$a^2+b^2=c^2$

$a^2+3^2=x^2$

$a=\sqrt{x^2-9}$

$$\ln|\sec\theta + \tan\theta| \Rightarrow \boxed{\ln\left|\frac{x}{3} + \frac{\sqrt{x^2+9}}{3}\right| + C}$$

7. [9 pts] Evaluate the Integral: $\int \frac{3x-1}{(x-2)(x+3)^2}dx$

$$\int \frac{3x-1}{(x-2)(x+3)^2} \Rightarrow \int \frac{A}{(x-2)} + \int \frac{B}{(x+3)} + \int \frac{C}{(x+3)^2}$$

$$\boxed{A\ LN|x-2| + B\ LN|x+3| + \frac{C}{(x+3)}}$$

why?

Copied wrong

show work!

$$\frac{\overset{3x-1}{3x+1}}{(x-2)(x+3)^2} = \frac{A}{(x-2)} + \frac{B}{(x+3)} + \frac{C}{(x+3)^2}$$

$$3x+1 = A(x+3)^2 + B(x+3)(x-2) + C(x-2)$$

$x=-3$ $\quad 3(-3)+1 = \cancel{A(-3+3)^2} + \cancel{B(-3+3)(-3-2)} + C(-3-2)$

$-8 = -5C$ $\quad \boxed{C = \frac{8}{5}}$

$x=2$ $\quad 3(2)+1 = A(2+3)^2 + \cancel{B(2+3)(2-2)} + \cancel{C(2-2)}$

$7 = 25A$ $\quad \boxed{A = \frac{25}{7}}$ $\quad A = 7/25$

$x=0$ $\quad 3(0)+1 = \frac{25}{7}(3)^2 + B(-6) + \frac{8}{5}\cdot(-2)$

$1 = \cancel{\frac{225}{7}} - 6B - \cancel{\frac{16}{5}}$

$-28.44 = -6B$

$\boxed{B = 4.8}$

+5

$$\boxed{\frac{25}{7} LN|x-2| + 4.8\ LN|x+3| + \frac{8}{5}\cdot\frac{1}{(x+3)^2} + C}$$

8. [4 pts] Evaluate the Integral: $\int \frac{3x-9}{x^2+4}dx$

$u = x^2+4$

$du = 2x\,du$

$\frac{du}{2x} = du$

$\int \frac{3x}{x^2+4} - \frac{9}{x^2+4}$

$\int \frac{3\cancel{x}}{4} \cdot \frac{du}{2\cancel{x}} \Rightarrow \frac{3}{2}\int \frac{1}{4}\,du = \frac{3}{2}\ln|x^2+4|$

$\int \frac{9}{x^2+4} \Rightarrow \left(\frac{9}{2}\right)\tan^{-1}\left(\frac{x}{2}\right)$

+4

$$\boxed{\frac{3}{2}\ln|x^2+4| - \left(\frac{9}{2}\right)\tan^{-1}\left(\frac{x}{2}\right) + C}$$

V. good work!

9. [4 pts] Evaluate the Integral: $\int \frac{x-2}{x-6}dx$

$x+6.$

$\int \frac{x-2+8-8}{x-6} \Rightarrow \int \frac{x-2+8}{x-6} - \int \frac{8}{x-6}$

$u = x-6$

$du = dx$

+1

$\int \frac{x-\cancel{6}}{\cancel{x}-6} - \int \frac{8}{u} \Rightarrow \int 1 - \int \frac{8}{4}$

$$\boxed{x - 8\ln|x-6| + C}$$

$u = (x^2+1)$ $\quad dv = \cos(x)$ $\quad u = x$ $\quad dv = \sin(x)$

① $du = 2x\,dx$ $\quad v = \sin(x)$ $\quad du = dx$ $\quad v = -\cos(x)$

$(x^2+1)\sin(x) - \int \sin(x) \cdot 2x\,dx$

$(x^2+1)\sin(x) - 2\int \sin(x) \cdot x\,dx$

$(x^2+1)\sin(x) - 2\left[-x\cos(x) - \int -\cos(x)\,dx\right]$

$(x^2+1)\sin(x) + 2x\cos(x) - 2\int \cos(x)\,dx$

$\boxed{(x^2+1)\sin(x) + 2x\cos(x) - 2\sin(x) + C}$

② $\int_0^1 32x e^{4x}$

$u = 32x$ $\quad dv = e^{4x}$ $\quad w = 4x$

$du = 32\,dx$ $\quad v = \frac{1}{4}\int e^{w} \cdot dw$ $\quad dw = 4\,dx$ $\quad \frac{dw}{4} = dx$

$v = \frac{1}{4}e^{4x}$

$32x\,\frac{1}{4}e^{4x} - \int \frac{1}{4}e^{4x} \cdot 32\,dx$

$32x\,\frac{1}{4}e^{4x} - 8\int e^{4x}\,dx$

$32x\,\frac{1}{4}e^{4x} - \frac{8}{4}e^{4x} \Rightarrow 32x\,\frac{1}{4}e^{4x} - 2e^{4x}$

-2

$\Big|_0^1 \left[32(1)\frac{1}{4}e^{4(1)} - 2e^{4(1)}\right] - \left[\cancel{32(0)\frac{1}{4}e^{4(0)}} - 2e^{4(0)}\right]$

$8e^4 - 2e^4 + 2 = \boxed{6e^4}$

(3) $\int \cot^3(x)\csc^5(x)\,dx$

$u = \csc(x)$
$du = -\csc(x)\cdot\cot(x)\,dx$
$\frac{du}{-\csc(x)\cot(x)} = dx$

$\cot^2 x + 1 = \csc^2 x$
$\cot^2 = \csc^2 x - 1$

$\int \cot^2(x)\csc^4(x)\cdot \cancel{\cot(x)}\cdot\cancel{\csc(x)}\cdot\frac{du}{-\cancel{\csc(x)\cot(x)}}$

$-\int \cot^2(x)\csc^4(x) \Rightarrow -\int (\csc^2 x - 1)\csc^4 x$

$-\int (u^2-1)\,u^4 \Rightarrow -\int u^6 - u^4$

$-\left[\frac{u^7}{7} - \frac{u^5}{5}\right] \Rightarrow \boxed{-\frac{\csc^7(x)}{7} + \frac{\csc^5(x)}{5} + C}$ ✓

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(4) $\int \tan^3(x)\,dx$

$\tan^2(x) + 1 = \sec^2(x)$
$\tan^2(x) = \sec^2(x) - 1$

$\int \tan^2(x)\tan(x)\,dx$

$\int (\sec^2(x) - 1)\tan(x)\,dx \Rightarrow \int \tan(x)\sec^2(x) - \tan(x)$

$\int \tan(x)\sec^2(x) - \int \tan(x)\,dx$ ✓

$u = \tan x$
$du = \sec^2(x)\,dx$
$\frac{du}{\sec^2(x)} = dx$

$\int u\cdot\cancel{\sec^2}(x)\cdot\frac{du}{\cancel{\sec^2}x} - \int \tan(x)\,dx$

$\int u\cdot du - \int \tan(x) \Rightarrow \frac{u^2}{2} + \ln|\cos x|$

$\boxed{\frac{\tan^2(x)}{2} + \ln|\cos x| + C}$ ✓